

**Data Mining**  
WS 2025/26

**Exercise Sheet 1: Clustering**

Submission until October 29th, 2025, 23:59

Discussion: October 30th, 2025

Please hand in your solutions as one jupyter notebook, where you clearly mark which snippet solves which exercise. Keep the order of the exercises.

**Exercise 1-1**      *Simple generation and visualization of data* (1 point)

- Create some data using python and sklearn, e.g., with *make\_blobs*<sup>1</sup>
- Visualize any two dimensions of the data in a 2d scatter plot, e.g., using *matplotlib*<sup>2</sup> or *seaborn*<sup>3</sup>.
- Visualize the data along its two main *principal components*<sup>4</sup> OR using a *tSNE* visualization<sup>5</sup>.

**Exercise 1-2**      *Different cluster algorithms solve different purposes* (1+1+2+1 points)

- a) Create data that contains at least three density-based clusters that cannot be detected correctly by k-Means<sup>6</sup>. Print the average NMI<sup>7</sup> of the clusters that k-Means finds over 10 runs (if you did everything correctly, it should be lower than 1.0).
- b) Create data that has at least four dimensions and contains at least four clusters that cannot be detected correctly by k-Means. Simultaneously, they should be detected correctly by DBSCAN<sup>8</sup>. Give the parameters with which DBSCAN can find those clusters. You may use the data generator DENSIRE<sup>9</sup>, but you don't have to.

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<sup>1</sup>[https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make\\_blobs.html](https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_blobs.html)

<sup>2</sup>[https://matplotlib.org/stable/api/\\_as\\_gen/matplotlib.pyplot.scatter.html](https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.scatter.html)

<sup>3</sup><https://seaborn.pydata.org/generated/seaborn.scatterplot.html>

<sup>4</sup>[https://scikit-learn.org/stable/auto\\_examples/decomposition/plot\\_pca\\_iris.html#sphx-glr-auto-examples-decomposition-plot-pca-iris-py](https://scikit-learn.org/stable/auto_examples/decomposition/plot_pca_iris.html#sphx-glr-auto-examples-decomposition-plot-pca-iris-py)

<sup>5</sup><https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html>

<sup>6</sup><https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

<sup>7</sup>[https://scikit-learn.org/stable/modules/generated/sklearn.metrics.normalized\\_mutual\\_info\\_score.html](https://scikit-learn.org/stable/modules/generated/sklearn.metrics.normalized_mutual_info_score.html)

<sup>8</sup><https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html#sklearn.cluster.DBSCAN>

<sup>9</sup><https://github.com/PhilJahn/DENSIRE>, described in Jahn, P., Frey, C. M., Beer, A., Leiber, C., & Seidl, T. (2024, August). Data with Density-Based Clusters: A Generator for Systematic Evaluation of Clustering Algorithms. In Joint European Conference on Machine Learning and Knowledge Discovery in Databases (pp. 3-21). Cham: Springer Nature Switzerland.

- c) Create data that has at least 10 dimensions and at least five clusters that cannot be detected correctly by DBSCAN. Simultaneously, they should be detected perfectly by HDBSCAN<sup>10</sup>. Can you detect the clusters with single linkage clustering<sup>11</sup>? Try it out! Explain why/why not!
- d) Regard the following properties for all three datasets from 1-2 a, b, and c:
- For each cluster, compute the average distance to its mean.
  - For each pair of clusters, compute the smallest distance between them.

Discuss the similarities and dissimilarities of the distributions.

### Exercise 1-3      *Evaluation*

(1+2+1 points)

- a) *For this subtask, use the output of sklearn's DBSCAN directly as an input for sklearn's NMI computation.* Create a data set with 3 clusters and find hyperparameters for DBSCAN s.t. the standard sklearn NMI implementation results in 1.0 even though DBSCAN only detects 2 clusters (and noise).
- b) Create a data set and two clusterings that show that NMI and ARI contradict each other. The clustering  $\mathcal{C}_1$  should be found by k-Means and the other clustering  $\mathcal{C}_2$  by DBSCAN. Contradicting means that
- $NMI(\mathcal{C}_1) > NMI(\mathcal{C}_2)$  and
  - $ARI(\mathcal{C}_1) < ARI(\mathcal{C}_2)$

Do **not** count noise (labeled by DBSCAN) as one cluster! Choose any other option and justify your decision.

- c) Implement the Silhouette Coefficient with the Manhattan distance instead of the Euclidean distance. To which of the following data transformations is the Manhattan version not invariant (but the Euclidean is)? Explain in your own words or give a small example.
- Translation (shifting all points by a constant vector)
  - Uniform scaling (multiplying all coordinates by a constant)
  - Rotation of the coordinate system
  - Reflection across an axis

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<sup>10</sup><https://scikit-learn.org/dev/modules/generated/sklearn.cluster.HDBSCAN.html>

<sup>11</sup><https://scikit-learn.org/dev/modules/generated/sklearn.cluster.AgglomerativeClustering.html>